THE QUANTUM STELLARATOR COOPERATION PROTOCOL

Governance Architecture, Logistics Framework, and Constitutional Foundations for a Planetary-Scale Quantum Laboratory

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Obstruction is productive. Every failure-to-glue contains the specification for its own resolution. The topology of what we cannot do here is the geometry of what must be done elsewhere.

The most important property of this protocol is not what it can compute but what it cannot express.

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Chapter 1

Preamble and Constitutional Foundations

1.1 Purpose

This document establishes the governance architecture, logistics framework, and constitutional foundations for the Quantum Stellarator Cooperation Protocol — a formally verified governance system for a seven-continent quantum laboratory network. The protocol governs the design, fabrication, and operation of a multi-layer topological quantum confinement device whose physical architecture requires planetary-scale cooperation and whose governance architecture is mathematically identical to its physics.

The protocol is designed to fail. Its resilience derives not from the prevention of failure but from the productive routing of failure through a cohomological structure that converts every obstruction into a specification for resolution. A network that has never failed has trivial topology and no protection. A network that fails well has the topology of a trefoil — irreducible, robust, and impossible to unknot by any smooth perturbation.

The entanglement between the device and the network that builds it is not metaphorical. The fabrication dependencies are non-local and non-separable. The geographic distances are the metric on the base space. The encrypted failure channel maintains useful superposition. The trefoil topology of the cooperation is the trefoil topology of the device. The geometry of the Earth is the geometry of the stellarator. Building the device and building the network are the same act.

1.2 Precedents

This protocol draws on four governance precedents, each of which solved a specific coordination problem at civilizational scale:

- **CERN (1954).** Demonstrated that multinational scientific infrastructure could be governed by a council with proportional contributions, open publication, and shared intellectual property. The key innovation was not the accelerator but the governance framework

that made multinational physics possible.

- **Antarctic Treaty (1959).** Demonstrated that a geographic commons could be maintained for 65+ years through the structural exclusion of military and sovereign territorial claims. Science-only access. The demilitarization is not enforced by power but by the structural incompatibility of military operations with treaty participation.
- **International Space Station (1998).** Demonstrated modular contributions from different nations that dock together via interface specifications. Each module is built to spec by a sovereign nation. Nobody owns the whole station. The interface specification *is* the governance.
- **Chile Neuro Rights Amendment (2021).** Demonstrated that constitutional constraints on technology can precede the technology's maturation. Cognitive liberty, mental privacy, and equitable access to neural augmentation were constitutionalized before the neurotechnology that would threaten them reached deployment. Governance preceded capability.

The pattern across all four: when the resource or the implications exceed any single institution's capacity, governance moves from contract to constitution. Not an agreement between parties, but a structural constraint that parties operate within.

1.3 The Seven Axioms

The protocol is founded on seven axioms. These axioms are not policy recommendations. They are formal constraints on the governance type system, verifiable by the same toolchain (Lean 4 / Cubical Agda) that verifies the device physics.

Axiom 1 (Conservation)**.** *Retraction preserves information. A reclassification from epistemic tier n to tier m produces four objects: the original claim, the reclassification record, the evidence that caused the reclassification, and the updated dependency graph. All four are conserved. Truth is reallocated, not destroyed.*

Axiom 2 (Symmetry)**.** *Promotion and demotion on the epistemic lattice are formally identical operations. Both trigger Rung Rule propagation through the dependency graph. Neither carries moral valence within the protocol. A node that reports a capability downgrade performs the same epistemic function as a node that reports a breakthrough.*

Axiom 3 (Fault as Type Error)**.** *When a dependency fails — an isotope is not delivered, a measurement contradicts a prediction, a fabrication run produces contaminated samples*

— the protocol does not ask who is responsible. It asks which type constraint was violated and what reallocation restores type safety. The compiler does not blame the programmer. It points to the line where the types do not match.

Axiom 4 (Load-Bearing Failure)**.** *A system requires nontrivial cohomology to be topologically protected. Obstructions are not defects in the network — they are the network's immune system. A collaboration that has never failed has never been tested. Designed failure pathways are structural investments in resilience, not costs to be minimized. A system with no obstructions has trivial cohomology, no topological protection, and no robustness against perturbation.*

Axiom 5 (Gauge Autonomy)**.** *Each node operates with full internal freedom. The protocol constrains only the transition functions — the interfaces between nodes. No node dictates another node's internal methodology, staffing, equipment choices, or institutional structure. Governance is a connection on overlaps, not a mandate on patches.*

Axiom 6 (Cryptographic Blame-Freedom)**.** *Failure reports are gauge-invariant: they contain the obstruction class (type, severity, routing specification) and no attribution. The encryption is not privacy protection — it is the mechanism that makes the failure channel trustworthy. A node that can report failure without consequence will report failure early. Early failure reporting produces computable cohomology. Computable cohomology produces optimal routing. Optimal routing produces resilience. The encryption is load-bearing infrastructure, not a courtesy.*

Axiom 7 (Cognitive Liberty DMZ)**.** *The governance type system has no constructors for cognitive coercion, weaponized application, or surveillance targeting. These operations are not prohibited — they are undefined. The governance sheaf has empty stalks in the DMZ. There is nothing to violate. The DMZ is a deliberate incompleteness in the type system: a region where power has no vocabulary. A system that can express everything can protect nothing. A system with a well-placed hole in its expressiveness protects everything the hole surrounds.*

Remark 1.1. Axioms 1–6 define how the protocol operates. Axiom 7 defines where the protocol *does not operate*. The DMZ is the boundary of the governance sheaf's support. Inside the support, the protocol computes, routes, and verifies. Outside the support, the protocol has no stalks, no sections, and no operations. The cognitive liberty DMZ is the most load-bearing structure in the protocol precisely because it is the structure where the protocol admits it has nothing to say.

Chapter 2

Mathematical Framework

2.1 The Governance Sheaf

Definition 2.1 (Base Manifold)**.** *Let M be the base manifold of the cooperation protocol. M is the geographic surface of the Earth, equipped with the standard Riemannian metric (geodesic distances between facilities). The seven continental nodes $\{U_\alpha\}_{\alpha\in\{1,\ldots,7\}}$ form an open cover of M.*

Definition 2.2 (Governance Sheaf)**.** *The governance sheaf $\mathcal{G}$ is a sheaf on M whose stalks at each point encode the capabilities, resources, epistemic commitments, and custody obligations of the nearest node. For each open set U_α:*

- *$\mathcal{G}(U_\alpha)$ is the set of local sections: the node's declared capabilities, resource inventory, and epistemic status for each architectural layer.*
- *The restriction maps $\rho_{\alpha\beta} : \mathcal{G}(U_\alpha \cap U_\beta) \to \mathcal{G}(U_\beta)$ encode the interface specifications between nodes — physical compatibility, epistemic compatibility, and custody transfer protocols.*

Definition 2.3 (Global Section)**.** *A global section $s \in \mathcal{G}(M)$ exists if and only if all local sections glue consistently across all overlaps:*

$$\rho_{\alpha\beta}(s|_{U_\alpha}) = \rho_{\beta\alpha}(s|_{U_\beta}) \quad \forall\, \alpha, \beta$$

The global section, when it exists, represents a complete, coherent device fabrication plan — every component, every custody chain, every interface satisfied.

Definition 2.4 (Obstruction Class)**.** *When local sections fail to glue, the failure is classified by a cohomology class $[\omega] \in H^1(M, \mathcal{G})$. The obstruction class encodes:*

1. *Type: which interface specification is violated (physical, epistemic, or custodial).*
2. *Severity: the magnitude of the incompatibility.*
3. *Routing specification: which nodes have the capability to resolve the obstruction.*

The obstruction class contains the complete information needed to route the failure to its resolution. It does not contain attribution.

2.2 Gauge Structure

Definition 2.5 (Gauge Group)**.** *The gauge group G is the group of local reference frame transformations at each node. Elements of G include: changes in internal methodology, equipment recalibration, personnel changes, institutional restructuring, and local regulatory compliance adaptations. The gauge group acts on local sections without changing the global state.*

Definition 2.6 (Connection)**.** *The governance connection ∇ is defined on the principal G-bundle over M. It specifies how to parallel-transport samples, data, epistemic claims, and custody records from one node to another. The connection is encoded in the custody and interface protocols.*

Definition 2.7 (Curvature)**.** *The curvature F_∇ of the governance connection measures the failure of parallel transport around a closed loop. In operational terms: when a sample, dataset, or custody record traverses a cycle of nodes and returns to its origin in a state inconsistent with its departure state, the curvature at that loop quantifies the inconsistency. Zero curvature means trivial cohomology, no topological protection, and fragile governance. Nonzero curvature means nontrivial cohomology, topological invariants, and robust governance.*

Theorem 2.1 (Productive Curvature)**.** *Let $\mathcal{N}$ be a network with governance connection ∇. If $F_\nabla = 0$ everywhere, then $H^1(M, \mathcal{G}) = 0$ and the network has no topological protection against perturbation. If $F_\nabla \neq 0$ at any loop, then $H^1(M, \mathcal{G}) \neq 0$ and the network possesses topological invariants that are stable under smooth deformation. The curvature is not a defect — it is the source of resilience.*

2.3 The Cognitive Liberty DMZ

Definition 2.8 (DMZ as Support Condition)**.** *Let $D \subset M \times \mathcal{A}$ be the cognitive liberty DMZ, where $\mathcal{A}$ is the application space (the set of all possible uses of the protocol's technology). The governance sheaf $\mathcal{G}$ satisfies the support condition:*

$$\mathcal{G}_p = \emptyset \quad \forall\, p \in D$$

The stalks of $\mathcal{G}$ are empty in D. No local sections exist. No operations are defined. The DMZ is not a zero section (which would require the stalk to exist). It is an absence of stalks entirely.

Proposition 2.1 (Inexpressibility)**.** *A governance operation targeting a point in D cannot be formulated within the protocol's type system. The operation is not prohibited (which would require a rule recognizing and rejecting it). It is undefined (the type system lacks the constructors to express it). Violation requires extending the sheaf into D, which requires modifying the constitutional axioms, which requires recompilation of all governance theorems, which is detectable by every node simultaneously.*

2.4 Encrypted Failure Routing

Definition 2.9 (Ring Signature Protocol)**.** *A failure report r from node α is signed with a ring signature σ_R over the ring $R = \{1, \ldots, 7\}$ of all continental nodes. The signature proves:*

1. *r was produced by a node in R (authentication).*
2. *r was not produced by any specific node (anonymity within the ring).*

The obstruction class $[\omega]$ extracted from r is gauge-invariant. The identity of α is gauge-dependent and is not transmitted.

Definition 2.10 (Zero-Knowledge Capability Proof)**.** *A node proves membership in a capability class C_k (e.g., "can provide millikelvin cryogenics within 90 days") without revealing which node is proving it. The proof is:*

$$ZKP\{(\alpha, c_\alpha) : c_\alpha \in C_k \wedge \alpha \in R\}$$

The network learns that some node can resolve the obstruction. The resolving node self-identifies only when it accepts the routing assignment.

Definition 2.11 (Superposition Routing)**.** *When a failure report is in anonymous superposition across all nodes, the DAO routing algorithm computes the optimal reallocation over all possible source-resolution pairs simultaneously. The superposition collapses only at the moment of resolution acceptance, when a specific node commits to providing the capability. Prior to collapse, the routing explores all pathways — analogous to quantum superposition enabling parallel evaluation of alternatives.*

Chapter 3

The Seven-Continent Laboratory Network

3.1 Design Principle: Cooperation by Physics

The seven-continent model is not a diplomatic aspiration. It is an engineering requirement. The device architecture demands materials, fabrication techniques, cryogenic infrastructure, isotope access, and verification expertise that are geographically distributed across every continent. No single continent possesses the complete capability set. The dependency graph is fully connected: every continent requires at least one resource or capability from every other continent. This interdependency is the enforcement mechanism. Cooperation is not negotiated — it is required by the physics.

3.2 Continental Node Assignments

3.2.1 Antarctica — Layer 0: The Governance Substrate

Antarctica contributes no fabrication capability and no materials. Its contribution is constitutional: the Antarctic Treaty (1959) provides the governance precedent that the entire protocol operates within. No territorial sovereignty, no military operations, science-only access, mandatory environmental stewardship.

Antarctic research stations (McMurdo, Concordia, Vostok) provide extreme cryogenic environments for field testing — ambient temperatures below $-80\,^{\circ}\mathrm{C}$. The Antarctic node has seasonal topology: access is periodic (approximately October–February), creating a time-dependent support for the local section. The protocol treats Antarctic seasonal closure identically to a node failure — the obstruction class routes governance functions to other nodes during the dark months, and the local section is restored when access resumes.

Primary role: Constitutional precedent, environmental stewardship protocol, extreme-environment testing.

3.2.2 Africa — Layers 0–1: Diamond, Helium, and Materials Gateway

Resource	Detail	TRL
Natural diamond	South Africa, Botswana; Element Six (De Beers) CVD growth	9
^{12}C substrates	Isotopically pure diamond feedstock from African geology	6–7
Helium-4	Rukwa basin (Tanzania), South African emerging production	4–5
SKA Observatory	Karoo, South Africa — precision scientific infrastructure	7–8

Table 3.1: African node resource inventory.

Africa provides the raw material foundation: carbon-12 for NV-center diamond substrates, emerging He-4 production, and the demonstrated capacity to host and operate precision scientific infrastructure at continental scale. The African node is the materials gateway — no NV sensing layer (layers 2–3) exists without African diamond supply chains.

Primary role: Diamond and isotopic carbon supply, emerging He-4 production, materials stewardship.

3.2.3 Asia — Layers 0–1: Nanofabrication and Graphene Patterning

Capability	Detail	TRL
EBL lithography	TSMC, Samsung, NIMS (Japan), KIST (Korea)	9
CVD graphene	Beijing Graphene Institute, Tsinghua, commercial	9
Twisted bilayer graphene	Multiple groups; magic-angle superlattice engineering	6
Quasiperiodic graphene	Penrose-patterned EBL — target capability	3

Table 3.2: Asian node capability inventory.

The semiconductor fabrication ecosystem is concentrated in East Asia. The quasiperiodic graphene patterning — the Penrose mask EBL work at sub-10 nm resolution — is most naturally performed where the world's lithographic capability lives.

Logistics constraint: CMOS-clean EBL facilities cannot process nickel-contaminated samples. All graphene patterning must be completed and samples sealed before any contact with Ni-containing components. The fabrication sequence is strictly ordered: graphene patterning (Asia) → characterization (Europe/North America) → Ni wire integration (separate metals facility, any continent).

Primary role: Graphene synthesis and quasiperiodic patterning, semiconductor-grade nanofabrication.

3.2.4 South America — Layers 0–1 and Constitutional Innovation

Resource/Capability	Detail	TRL
Niobium	CBMM (Brazil) — ∼80% of global Nb supply	9
Neuro rights framework	Chile constitutional amendment (2021)	8
Constitutional XAI context	Legal precedent for pre-deployment governance	6

Table 3.3: South American node resource and governance inventory.

South America provides both the foundational superconducting material (Brazilian niobium — Layer 0) and the constitutional innovation (Chilean neuro rights — Axiom 7). CBMM's near-monopoly on global Nb production makes South American participation structurally mandatory for the superconducting substrate. Chile's neuro rights amendment provides the legal precedent for the cognitive liberty DMZ — governance preceding capability.

Primary role: Niobium supply, constitutional governance innovation, cognitive liberty framework.

3.2.5 Europe — Layers 4–5: Superfluid Physics and Stellarator Engineering

Capability	Detail	TRL
Stellarator engineering	W7-X (Greifswald); VMEC/SIMSOPT codes	8
Millikelvin cryogenics	Lancaster, Aalto (Finland), Néel (Grenoble)	9
He-3 superfluid physics	Lancaster, Aalto — world-leading programs	8
Analog gravity experiments	Multiple European groups	5
CERN governance model	Multinational laboratory governance precedent	9

Table 3.4: European node capability inventory.

Europe is the natural home for layers 4–5. Wendelstein 7-X provides stellarator coil optimization expertise. Lancaster and Aalto provide the world's leading He-3 superfluid experimental programs. CERN provides the organizational template for multinational laboratory governance.

Logistics constraint: He-3 experiments require millikelvin temperatures achieved via dilution refrigerators. He-3 is custodially loaned from the North American DOE system — the

European node cannot operate layers 4–5 without the North American custodial relationship. This dependency is by design.

Primary role: Superfluid physics, stellarator engineering, CERN-model governance.

3.2.6 North America — Layers 6–7: Verification, Isotopes, and Control

Capability	Detail	TRL
Isotope enrichment	ORNL Stable Isotope Production & Research Center	7–8
He-3 custodial system	DOE national laboratory infrastructure	8
GoS framework	Lean 4 formal verification (this project)	5
Quantum simulation	NVIDIA CUDA-Q, Microsoft Station Q, IBM Quantum	7–8
Formal methods community	CMU, MIT, Stanford, Microsoft Research	9

Table 3.5: North American node capability inventory.

North America holds isotope enrichment capability (Ni-62 at ORNL), the He-3 custodial system (DOE), the formal verification framework (GoS), and the quantum computing simulation ecosystem (CUDA-Q, Station Q). The ZX calculus compilation and OAM control system development (layers 6–7) sit here alongside the verification tools that check the entire architecture.

Logistics constraint: Ni-62 enrichment is performed at national laboratory facilities under DOE oversight. Enriched material is tracked and must be returned after use. The isotope custody model operates identically to the He-3 custodial loan system.

Primary role: Isotope enrichment and custody, formal verification, quantum simulation, ZX/OAM control layer.

3.2.7 Oceania — The Integration Topos

Capability	Detail	TRL
Quantum computing labs	UNSW Sydney, University of Sydney, ARC CoE	7
Broad measurement capability	Cryo, electronic, optical, computational	6
Pacific network position	Geographic hub between Asia and Americas	—

Table 3.6: Oceanian node capability inventory.

Oceania serves as the integration topos — the node where characterized components from every other continent are received, assembled, tested, and validated as a composed device.

The integration lab requires the broadest measurement capability but not the deepest specialization in any single domain. Australia's quantum ecosystem has exactly that profile. Geographic position between Asia and the Americas makes Oceania a natural hub for Pacific-spanning logistics.

Primary role: Multi-layer integration, composed device testing, network coordination.

3.3 Dependency Graph

The network's dependency structure is fully connected. Figure 3.7 shows the critical material and capability dependencies between continental nodes.

From	**To**	**Dependency**
South America	All	Niobium (Layer 0 substrate)
Africa	Asia, Europe	^{12}C diamond substrates for NV centers
Africa	Europe, N. America	He-4 supply
Asia	Europe, Oceania	Patterned graphene samples
N. America	Europe	He-3 custodial loan for superfluid experiments
N. America	All	Ni-62 enriched isotope supply
N. America	All	GoS formal verification of all layers
Europe	Oceania	Characterized superfluid components
Europe	All	Stellarator coil optimization codes
All	Oceania	Characterized layer components for integration
Antarctica	All	Constitutional governance precedent
S. America	All	Cognitive liberty / neuro rights framework

Table 3.7: Critical dependencies between continental nodes. No node is self-sufficient.

Chapter 4

Logistics Framework

4.1 The Isotope Custody Model

4.1.1 Principle: Custody, Not Ownership

No node in the network owns enriched isotopic materials. All enriched isotopes are held in custody under protocols modeled on the DOE He-3 custodial loan system. The custody model has three tiers:

1. **Sovereign custody.** The enrichment facility (e.g., ORNL for Ni-62, DOE for He-3) retains ultimate ownership. Material is loaned to research nodes under custodial agreement.
2. **Operational custody.** The research node holding the material for active experimental use. Operational custody includes responsibility for containment, accounting, and recovery.
3. **Transit custody.** The logistics chain during material transfer between nodes. Transit custody includes responsibility for maintaining cryogenic conditions (for He), contamination prevention (for Ni-62), and chain-of-custody documentation.

All custody transfers are recorded as state transitions in the DAO. Each transition is a formally verified operation on the governance sheaf — it either type-checks or it does not.

4.1.2 Design for Recovery

Every device assembly that incorporates an enriched isotope must satisfy a recovery constraint: a documented, physically tested disassembly procedure that recovers the isotope with $> 90\%$ yield. This constraint is enforced at the design stage, not at decommissioning.

Definition 4.1 (Recovery Pathway)**.** *A recovery pathway for isotope I in assembly A is a*

verified procedure P_I^A such that:

$$yield(P_I^A) > 0.90$$

P_I^A is tested on a non-enriched surrogate before enriched material is committed.

No enriched isotope may be incorporated into an assembly without a verified recovery pathway.

4.1.3 Isotope Inventory

Isotope	Layer	Source	Custody Model	Recovery
^{93}Nb	0	CBMM (Brazil)	Commercial purchase	Melt/recast
^{12}C	2–3	Cambridge Isotope / Element Six	Purchase; recoverable	Mechanical separation
^{62}Ni	4–5	ORNL enrichment	DOE custodial loan	Wire extraction
^{4}He	4–5	Commercial (crisis pricing)	Purchase + recycling	Closed-loop recovery
^{3}He	4–5	DOE	Custodial loan; return mandatory	Sealed recovery system
^{209}Bi	6–7	Commercial (100% natural)	Purchase	Standard recovery

Table 4.1: Isotope inventory with custody models and recovery pathways.

4.1.4 Niobium: The Exception

Niobium (^{93}Nb) is 100% naturally abundant and does not require isotope enrichment. It is commercially available from CBMM (Brazil), H.C. Starck, and Plansee. However, CBMM's ∼80% share of global Nb production creates a single-source dependency that the protocol must address through strategic stockpiling and supply diversification. Nb is the only material in the architecture that is purchased outright rather than held in custody.

4.2 Cleanroom Segregation Protocol

4.2.1 The Nickel Contamination Constraint

Nickel is classified as a non-CMOS-compatible material in all semiconductor cleanroom protocols. Nickel ions are mobile in semiconductor materials and cause device failure at contam-

ination levels above 10^{10} atoms/cm^2. Any sample that contacts nickel-contaminated tooling is permanently barred from CMOS-clean processing.

This creates a hard fabrication sequence constraint: the graphene patterning (CMOS-clean EBL) must be completed and samples sealed in contamination-free packaging *before* any Ni-62 wire fabrication, handling, or integration occurs. The two processes cannot share a facility, cannot share tools, and cannot share a cleanroom.

4.2.2 Fabrication Sequence

The mandated fabrication sequence is:

1. **Stage 1 — CMOS-clean (Asia).** Graphene CVD growth, Penrose mask EBL patterning, graphene transfer to Nb substrate. All operations in CMOS-clean or semi-clean classified tools. Output: sealed, characterized graphene/Nb heterostructure.
2. **Stage 2 — Diamond integration (Africa/Asia).** NV-center nanodiamond deposition on graphene surface. CMOS-compatible process. Output: sealed graphene/Nb/NV-diamond stack.
3. **Stage 3 — Metals fabrication (North America).** Ni-62 trefoil wire winding in a metals-classified facility. Completely separate from Stages 1–2. Output: characterized Ni-62 trefoil coil assembly.
4. **Stage 4 — Superfluid cell (Europe).** He-3/He-4 cell fabrication and cryogenic testing. Independent of Stages 1–3. Output: characterized superfluid confinement cell.
5. **Stage 5 — Control layer (North America).** Bi thin film deposition, OAM optical assembly. Can proceed in parallel with Stages 1–4. Output: characterized Bi/OAM control module.
6. **Stage 6 — Integration (Oceania).** All sealed, characterized components are received and assembled into the stellarator geometry. First contact between CMOS-clean and metals components occurs only at this stage, in a controlled integration environment (not a CMOS cleanroom). Output: assembled device ready for cryogenic testing.

Critical rule: The flow from CMOS-clean to metals is one-directional and irreversible. No component that has entered the metals pathway may return to CMOS-clean processing. The governance sheaf encodes this as a one-way restriction map: sections can be restricted from clean to metals but not lifted from metals to clean.

4.3 Helium Supply Chain

4.3.1 He-4: Commercial Market Under Stress

The global He-4 market is in crisis as of March 2026. Qatar's Ras Laffan facility, producing approximately 30% of global supply, has been disrupted by the closure of the Strait of Hormuz. Spot prices have surged 70–100%. The semiconductor industry (17% of US helium demand) competes directly with the quantum research community for the same supply.

The protocol mandates closed-loop He-4 recovery for all cryogenic operations. The MRI industry has demonstrated that helium recycling systems can reduce consumption by up to 80%. Every node operating cryogenic equipment must implement comparable recovery infrastructure.

Supply diversification: The protocol sources He-4 from multiple geographic origins — US (Shute Creek, Wyoming), Qatar (when available), emerging African production (Tanzania Rukwa basin), and Canadian/Australian projects in development. No single source may exceed 40% of the network's total He-4 consumption.

4.3.2 He-3: Strategic Material Under DOE Control

He-3 is a byproduct of tritium decay from nuclear weapons programs. Global supply is estimated at approximately 15,000–20,000 liters per year. The DOE monitors all US He-3 distribution. Research allocations compete with neutron detection for nuclear security.

The protocol operates He-3 under the existing DOE custodial framework:

- All He-3 is loaned, not purchased.
- Custodial agreements specify return obligations.
- Every cryostat using He-3 must have a sealed recovery system with documented boiloff rate.
- He-3 losses during experimental runs are reported as custody adjustments in the DAO — not as failures (Axiom 6 applies; reporting is blame-free).

4.3.3 Existing Industry Model: MRI Helium Management

The medical imaging industry manages helium at a scale directly relevant to this protocol. Approximately 60,000 MRI machines worldwide each require 1,500–2,000 liters of He-4 for su-

perconducting magnet cooling. The industry has developed long-term supply contracts (10+ years), closed-loop recycling systems, and logistics infrastructure for cryogenic transport.

The protocol leverages this existing infrastructure rather than building parallel systems. Where a continental node is co-located with MRI helium management infrastructure (e.g., Philips/Air Products supply chains in Europe and North America), the node uses the same logistics providers, the same cryogenic transport containers, and the same recovery systems.

4.4 Cryogenic Transport Logistics

4.4.1 The Boiloff Constraint

Liquid helium continuously evaporates during transport. Standard cryogenic ISO containers maintain liquid He-4 at −269 °C with a boiloff rate of approximately 0.5–1.0% per day. A shipment from Qatar to East Asia via the Strait of Hormuz (pre-crisis: 10–14 days) arrives with 85–93% of its original volume. Via the Cape of Good Hope (24–28 days), arrival volume drops to 72–86%. The routing decision is not just logistical — it determines how much material arrives.

The governance sheaf encodes boiloff as a metric property of the base manifold: the effective distance between two nodes is not the geodesic distance on the Earth's surface but the *cryogenic effective distance* — geodesic distance scaled by the boiloff rate. Two nodes that are geographically close but lack cryogenic transport infrastructure may be effectively farther apart than two nodes on opposite sides of the planet with established supply chains.

4.4.2 Thermal Staging Architecture

The device operates across six orders of magnitude in temperature. No single cryogenic system spans this range. The thermal staging architecture requires concentric thermal shields:

Stage	Temperature	Layers	Cooling Technology
Room temp	∼300 K	OAM optics, external control	Ambient
LN_2 shield	77 K	Radiation shielding	Liquid nitrogen
LHe shield	4.2 K	Bi surface states, Nb substrate	Liquid He-4 bath
He-4 pot	1.5 K	He-4 superfluid	Pumped He-4
Still	∼0.7 K	Thermal anchor	Dilution refrigerator
Mixing chamber	5–10 mK	He-3 superfluid, Ni-62 wire	Dilution refrigerator

Table 4.2: Thermal staging architecture.

The trefoil geometry adds a constraint that no existing cryostat design accommodates: the sample space is topologically nontrivial. Standard dilution refrigerators have cylindrical or rectangular sample spaces with trivial topology. A trefoil-shaped sample space requires custom cryostat design — an engineering task that is itself at TRL 1–2.

4.5 Geopolitical Supply Chain Risks

Material	**Chokepoint**	**Severity**	**Mitigation**
He-4	Strait of Hormuz (Qatar)	Critical	Diversify to US/Tanzania/Australia
He-3	DOE policy / tritium program	High	European reactor production (ILL)
Nb	CBMM market dominance	Medium	Strategic stockpile; Australian alt.
Ni-62	ORNL availability	Medium	Russian alt. sources (sanctions risk)
^{12}C	Cambridge Isotope supply	Low	Multiple suppliers; synthetic routes
Bi	Widely distributed	Low	No chokepoint

Table 4.3: Geopolitical supply chain risk assessment.

Chapter 5

DAO Execution Model

5.1 Architecture

The Decentralized Autonomous Organization (DAO) is the execution layer of the governance protocol. It translates the formally verified constitutional axioms into executable operations. The DAO has three functional layers:

1. **Verification layer.** Lean 4 / Cubical Agda theorems encoding the seven axioms, the custody protocols, the failure-routing rules, and the epistemic status propagation. These are the source of truth.
2. **Governance layer.** Smart contracts compiled from the verified theorems. Each contract implements a specific governance operation: custody transfer, failure report processing, obstruction routing, capability proof verification, or DMZ boundary enforcement. The compilation from theorem to contract is itself verified.
3. **Execution layer.** The runtime environment where contracts execute. State transitions are transparent, recorded, and auditable by all nodes. Treasury management (isotope custody pool) is algorithmic.

5.2 Consensus: Proof-of-Proof

The DAO's consensus mechanism is proof-of-proof. Governance weight is earned through verified contributions, not through resource expenditure (proof-of-work) or capital accumulation (proof-of-stake).

Definition 5.1 (Governance Weight)**.** *Each node's governance weight w_α is a function of two orthogonal dimensions:*

1. ***Epistemic weight*** w_α^E*: proportional to the node's verified epistemic contributions (Proved-tier theorems, Demonstrated-tier experimental results, Motivated-tier theoretical analyses). Higher-tier contributions carry more weight.*

2. ***Resilience weight*** w_α^R*: proportional to the node's demonstrated capacity to process failures. A node that has successfully routed obstructions — reported failures early, absorbed rerouted work, maintained operations during supply disruptions — earns resilience weight regardless of its publication record.*

Total governance weight: $w_\alpha = w_\alpha^E + w_\alpha^R$*. The two dimensions are orthogonal: a node earns standing by producing truth or by processing failure productively. Both are load-bearing.*

5.3 State Transitions

All governance operations are state transitions on the governance sheaf. Each transition has a formally specified type:

Transition	Type Signature	Description
`custody_transfer`	Node × Isotope × Node → CustodyState	Transfer isotope custody between nodes
`failure_report`	ObstructionClass → RoutingPlan	Anonymous failure → resolution routing
`capability_proof`	Node × Capability → ZKP	Prove capability without revealing node
`epistemic_update`	Claim × Tier × Tier → DepGraph	Reclassify claim; propagate Rung Rule
`recovery_verify`	Isotope × Assembly → RecoveryPath	Verify recovery pathway before commitment
`dmz_check`	Operation → Bool	Verify operation is within sheaf support

Table 5.1: DAO state transition type signatures.

Every transition is type-checked against the constitutional axioms before execution. A `custody_transfer` that would route enriched material to a node without a verified recovery pathway fails the type check and is rejected. A `failure_report` that contains attribution fails the gauge-invariance check and is rejected. A `dmz_check` that returns `false` halts the operation — the governance sheaf has no stalk at that point in application space.

5.4 Treasury: The Isotope Custody Pool

The DAO treasury is not a financial pool. It is a custody ledger: a formally verified record of which isotopes are held by which nodes under which custodial agreements, with which recovery pathway commitments and which return obligations.

The treasury is managed by the following invariants:

- **Conservation:** Total isotope mass across all custodial states is tracked. Losses (boiloff, contamination, experimental consumption) are recorded as custody adjustments, not hidden.
- **Recovery obligation:** Every custody assignment carries a recovery commitment. The commitment is verified before the assignment is executed.
- **No bottleneck:** No single node may hold more than 40% of the network's operational custody of any single isotope. Concentration triggers automatic redistribution.
- **Sovereign return:** All enriched isotopes return to their sovereign custodian (ORNL, DOE, etc.) upon network decommissioning. The DAO cannot override sovereign custody.

Chapter 6

Integration with the GQAC Framework

6.1 Scope

This Cooperation Protocol is designed to integrate with the Global Quantum & AI Commons (GQAC) framework as the formally verified operational core. The seven axioms, the sheaf-cohomological governance structure, the DAO execution model, and the cognitive liberty DMZ constitute the protocol's contribution to the GQAC. Integration details, including alignment with existing GQAC governance structures and operational procedures, are deferred to a joint document.

6.2 Three-Topos Architecture

The complete system operates across three formally verified topoi:

1. **Physics topos.** The device layer stack, topological invariants, and Glassbox epistemic tiers. Verified in Lean 4 / Cubical Agda. Governs what is physically true.
2. **Governance topos.** The seven axioms, custody protocols, failure-routing rules, and epistemic status propagation. Verified by the same toolchain. Governs what is operationally permitted.
3. **Interface topos.** The bridge between physics and governance. Each material dependency has both a physics type (what it does in the device) and a governance type (who holds it, under what terms, with what obligations). The interface topos verifies that the physics and governance are mutually consistent.

All three topoi are checked by the same compiler. The GoS repository becomes a repository not just for physics theorems but for governance theorems, verified with the same rigor and the same tools.

6.3 The Compiler Is the Credential

The protocol does not require institutional authority. It requires verified axioms. Any individual or institution can propose a governance theorem, a custody protocol modification, or a new capability proof. The proposal is accepted if and only if it type-checks against the constitutional axioms and passes verification.

The author of this protocol holds no institutional affiliation. The protocol's authority derives from the same source as the GoS framework's 144 theorems: the compiler accepts or rejects without regard to who wrote the code. This principle extends from physics verification to governance verification. The compiler is the credential for both.

End of Document

Draft — March 2026

Quantum Stellarator Cooperation Protocol v0.1

This protocol is designed to fail.

Its resilience is its topology.